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# A watermarking-based authentication and recovery scheme for encrypted audio

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**Abstract**
To protect the audio stored on third-party servers, a novel authentication and recovery watermarking scheme for encrypted audio is proposed. We define the relative energy (RE) feature and analyze the characteristic of the feature. At the embedding stage, the host audio is encrypted firstly. Then the encrypted audio is divided into frames. We embed watermark bits into each frame by quantifying the RE feature. At the decoding stage, the receivers locate the attacked frames based on the watermark extraction and substitute the attacked frames using 0 amplitude signals, which are scattered over different segments after anti-scrambling transformation and do not influence the expressed meaning of watermarked signal. Experimental evaluation results demonstrate that the scheme improves the security of audio signals stored on third-party servers.

**Keywords** Digital watermark · Encrypted audio · Content authentication · Tamper recovery

## 1 Introduction

In recent years, users are increasingly storing their data on third-party servers, e.g., cloud storage servers, which bring growing convenience to data accessing. In this case, users can move their data onto cloud servers and don't have to care about the complexities of direct hardware management [13]. However, for the users or data owners, the third-party servers are not within their trusted domains and the sensitive data are usually encrypted before uploading onto third-party servers. It is possible that the encrypted data downloaded from

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third-party servers have been tampered, occurred on these servers or during the process of transmission. The data loss incidents of third-party servers appear from time to time [14].

If the digital audios stored on third-party servers are attacked, there will be two issues that need attention. First, if users act accordingly based on the attacked signal, it may cause serious consequences. Second, the decryption process could be very time consuming. If a large amount of the encrypted data is attacked, decryption of the data will take a lot of useless time. Fortunately, authenticating the encrypted signal before decryption can solve the two issues simultaneously.

Digital watermarking technology has received increasing attention in recent years, and many watermarking algorithms have been proposed. According to different application scenarios of multimedia files, digital watermarking can be divided into fragile watermarking and robust watermarking. In most situations, fragile watermarking is for integrity authentication [7, 18], while robust watermarking is for copyright protection [1].

As one kind of audio watermarking used for integrity authentication, at the embedding stage, users divide the host audio into some frames. Then they calculate the fragile feature of each frame and embed watermark bits into the frame by quantifying the feature. At the decoding stage, the receivers calculate the feature of watermarked audio and extract watermark bits. If the extracted watermark bit is the same as the original one, it indicates that the audio is intact. Otherwise, the audio is attacked. In [6], Liu *et al*. proposed an authentication and recovery algorithm for speech signal based on digital watermarking. The compression method for digital speech was discussed and one block-based large capacity embedding method was explored. In this paper, compressed signal as watermark was embedded into host signal, used for the reconstruction of attacked signal. Experimental results demonstrate that the scheme proposed in [6] improves the security and the accuracy of tamper location, and has a certain ability of tamper recovery for speech signals.

As for robust audio watermarking, many watermarking methods have been developed. In [15], Wang *et al.* proposed a robust audio watermarking method based on invariant exponent moments and synchronization code technique. The scheme was resistance against some attacks, based on the principle that the binary image extracted from watermarked signal after being attacked was similar to the original one. In [8], Natgunanathan *et al*. proposed a multilayer watermarking algorithm based on the patchwork method. In this algorithm, authors embedded watermark in multiple layers in any order. In addition, embedding watermark in one layer does not affect the watermark embedded in other layers, experimental results shown that the proposed method can withstand a wide range of common attacks.

In recently, some researchers have proposed several robust reversible watermarking algorithms, called cover-lossless robust watermarking techniques, such as audio-lossless robust watermarking methods, image-lossless robust watermarking. Audio-lossless robust watermarking can resist those content-preserving manipulations while can restore the cover audio files completely in case of no attacks. There are many ALRW methods proposed focus on additive noise-like manipulations such as MP3 compression, low-pass filtering and Gaussian additive noise [4, 16, 17]. While there are few ALRW countermeasures proposed in the literature that are resistant to challenging de-synchronization attacks such as random cropping and time-scale modification (TSM). The main reason is that the exploited robust features in the existing methods are related to the sample position. In [3], Li *et al*. proposed a novel audio-lossless robust watermarking against de-synchronization attacks. The method can reduce the compensation information and provide strong robustness to TSM. In case of no attacks, the cover audio signal can be recovered without any loss.

Although digital watermarking has a large number of research results, the above-mentioned algorithms with superior performance are useful for the audio signals that do not require encryption. The methods do not provide good protection for encrypted audio signals stored in third parties and cannot easily be directly applied to audio signals. So, relatively few audio watermarking algorithms are available for the encrypted domain.

Considering the background and motivation above, a tamper recovery scheme for digital audio in encrypted domain based on digital watermark is proposed. Firstly, the relative energy (RE) feature is defined and the properties are analyzed. Then an encryption method is given, containing scrambling and multiplication. It's hard for attackers to get the meaningful content from encrypted signal. And the method can protect the privacy of sensitive audio signals. In this paper, a pseudo-random sequence is divided into different sets, and each set is mapped onto the corresponding frame number. The set and frame number as watermark are embedded by quantifying the RE feature. The set as the identification of each frame is used to locate the watermarked and attacked frames. Based on the precise tamper location, the attacked frames are substituted by using 0 amplitude signals, which will be spread to different parts after anti-scrambling and not influence the meaning of the original signal. Theoretical analysis and experimental results demonstrate that, the encryption method improves the security of audio signals stored on third-party servers.

The organization of this paper is as follows. Section 2 introduces the relative energy feature, including the definition and properties. Section 3 describes the scheme proposed. Section 4 analyzes the performance of the algorithm and shows the simulation results. Finally, we summarize the conclusion in Section 5.

## 2 Relative energy

Signal energy as a feature has been widely used for watermark embedding [2, 10, 12]. Those existing schemes are vulnerable to substitution attack [5], for the feature is public. In the following subsections, the definition of signal energy ratio is given, and a new watermark embedding method is proposed based on signal energy ratio, in order to increase the security of watermark embedding scheme.

### 2.1 Definition of relative energy

We denote the two $N$ length different audio as $X=\{x(i), 1\le i\le N\}$ and $Y=\{y(i), 1\le i\le N\}$, where $x(i)$ and $y(i)$ represent the $i$-th sample. We define the relative energy (RE) of $X$ to $Y$ as

$$RE(X,Y) = 10\lg\left(\sum_{i=1}^{N}\left(\frac{x(i)}{\alpha + y(i)}\right)^2\right) \tag{1}$$

where $\alpha > 0$.

We analyze the characteristics of the RE features by the following experiment. We select three fragments 1000 length audio, denoted by $Y1$, $Y2$ and $Y3$ (shown in Figs. 1, 2 and 3). The energy of $Y1$, $Y2$ and $Y3$ are 72.18, 99.29 and 18.75, respectively. And then we select one 100,000 length audio (shown in Fig. 4) and divide the audio into 100 frames. The energy of the 100 frames is shown in Fig. 5. Based on Eq. (1), we calculate the RE feature

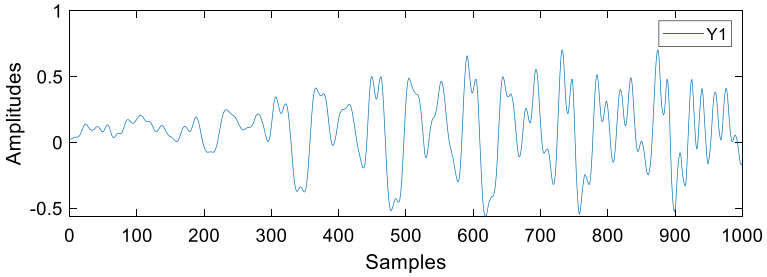



**Fig. 1** The 1000 length audio $Y1$

of each frame of $X$ to $Y1$, $Y2$ and $Y3$. The calculated RE features are shown in Fig. 6. The results demonstrate that:

1) Under the condition that the signal $Y$ is invariable, the larger the energy of $X$, the larger the RE feature. It means that a higher feature indicates a larger energy of $X$.
2) The RE feature of each frame of $X$ to $Y1$ are different. And the RE features of the same frame of $X$ to $Y1$, $Y2$ and $Y3$ are different too. It indicates that, for a given RE feature and the signal $X$, it is difficult to get the signal $Y$. So, to get the RE feature of $X$ to $Y$, both $X$ and $Y$ are needed. In this paper, we use the signal $Y$ as the key, to improve the security of the watermarking system.

## 2.2 Relationship between Samples and Relative Energy

The primary goal of digital watermarking is to embed some information, used for copyright protection or content forensics, into a host audio. When necessary, users can extract the information to prove the ownership of the audio or identify the authenticity of the signal. In the embedding stage, the goal is to embed the information (called watermark) into the host audio ($X$). First, users calculate the feature (RE). According to the information, users quantify the feature RE to $Q_{RE}$. Secondly, based on the quantified feature $Q_{RE}$, they can generate the watermarked audio ($X'$).

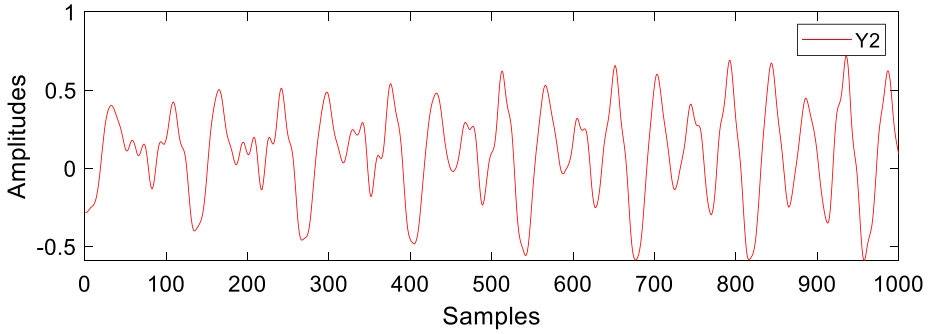



**Fig. 2** The 1000 length audio $Y2$

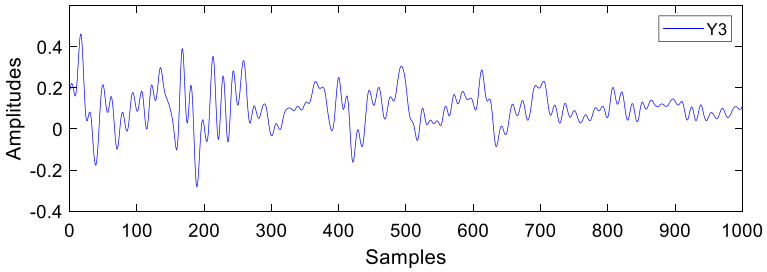



**Fig. 3** The 1000 length audio $Y3$

According to the strategy of watermark embedding, we given the method for generating watermarked audio according to quantified feature. Using the same symbols as above, $X$ denotes the host audio, $Y$ denotes the $N$ length audio different to $X$, $RE$ denotes the feature of host audio, $Q_{RE}$ denotes the quantified feature, and $X^{'} = \{x^{'}(i), 1 \le i \le N\}$ denotes watermarked audio, where $x^{'}(i)$ is the $i$-th sample.

According the Eq. (1), we get

$$\sum_{i=1}^{N} \left( \frac{x(i)}{\alpha + y(i)} \right)^2 = 10^{\frac{E}{10}} \tag{2}$$

$$\sum_{i=1}^{N} \left( \frac{x'(i)}{\alpha + y(i)} \right)^2 = 10^{\frac{Q_{RE}}{10}} \tag{3}$$

Combining the Eqs. (2) and Eq. (3) we have

$$\frac{\sum_{i=1}^{N} \left( \frac{x'(i)}{\alpha + y(i)} \right)^2}{\sum_{i=1}^{N} \left( \frac{x(i)}{\alpha + y(i)} \right)^2} = 10^{\frac{Q_{RE} - RE}{10}} \tag{4}$$

In the embedding stage, the primary goal is to generate the watermarked signal ($X^{'}$ in this section). In the Eq. (4), $x^{'}(i)$ is unknown, and all other parameters are known. The process of obtaining $x^{'}(i)$, is the process of solving Eq. (4). Based on the above analysis, we can get that the one of the solutions can be represented by the Eq. (5). So, based on the Eq. (5), we can get the watermarked audio according to the quantized feature $Q_{RE}$.

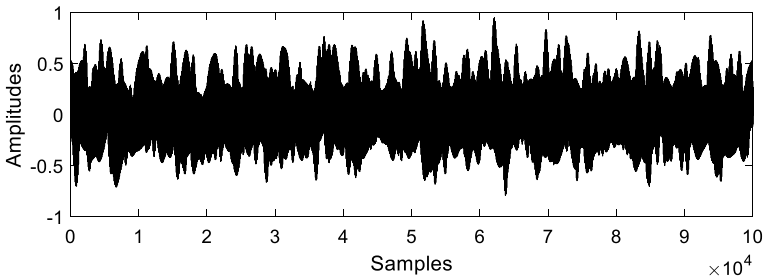



**Fig. 4** The 100,000 length audio $X$

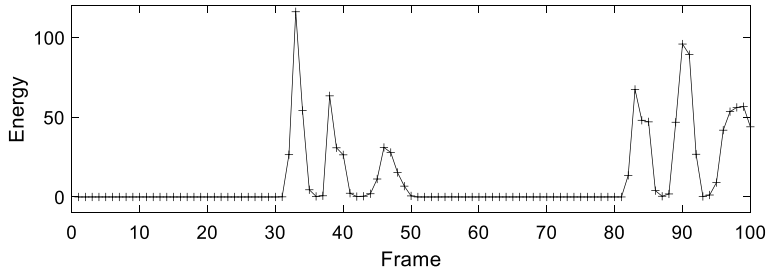



**Fig. 5** The energy of 100 frames

$$x'(i) = x(i)\sqrt{10^{\frac{Q_{RE}-RE}{10}}} \tag{5}$$

## 3 The scheme

There are two main methods for authenticating encrypted audio. The first is to decrypt the encrypted audio, and then authenticate decrypted signal. In this case, if the encrypted audio has been tampered, the operation of decryption will not generate correct results. Especially for a large amount of audio attacked, the decryption process is time consuming. The second method is to authenticate the encrypted audio before employing the decryption process. If signal is attacked, they will be discarded and the decryption process will not be performed. On the other hand, if the signal is intact, the decryption process will be performed, and the original audio signals can be obtained. Thus, authenticating the encrypted audio signals directly not only can verify the authenticity of the audio, but also improve the efficiency. For the scheme proposed in this paper, watermark bits are embedded into host audio in encrypted domain, and the authentication process is performed on the encrypted signals before decryption. By doing this, authentication and decryption efficiencies are improved.

The main work of the scheme proposed is shown in the dotted rectangle of Fig. 7, focusing on watermark embedding and content authentication for encrypted data.

### 3.1 Watermark Embedding

We suppose $X$ as the $L$ length audio, $X = \{x_l, 1 \le l \le L\}$, where $x_l$ represents the $l$-th sample.

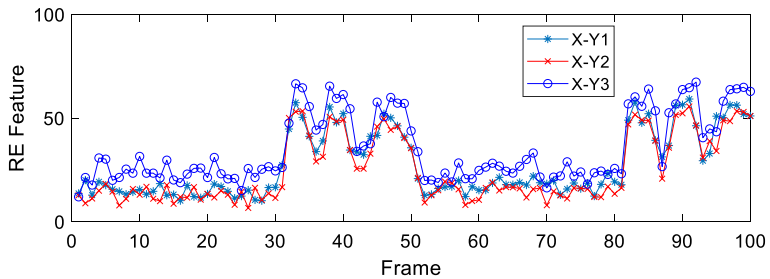



**Fig. 6** The RE feature of each frame of $X$ to $Y1$, $Y2$ and $Y3$

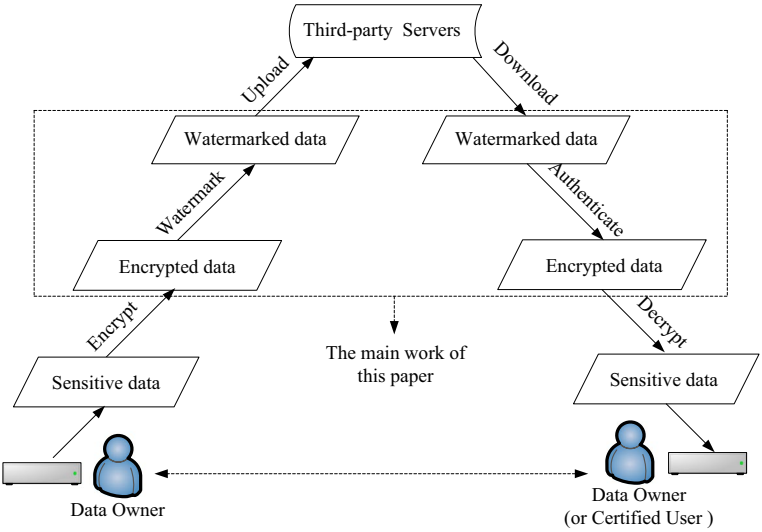



**Fig. 7** Main work of the scheme

### 3.1.1 Preprocessing

① Scrambling: the host audio $X$ is scrambled, by using the method based on logistic chaotic mapping, which is shown in Eq. (6).

$$y_{l+1} = \mu y_l(1 - y_l), y_0 = k \tag{6}$$

where $k$ is the initial value of the logistic map, as the key of the watermarking system. $\mu$ is the logistic parameter, $0 \le \mu \le 4$. Under the condition that $3.5699 \le \mu \le 4$, the sequence generated by Eq. (6) is a pseudo-random distribution state, especially when the value of $\mu$ is close to 4.

Denote $Y = \{y_l | l = 1, 2, \ldots L\}$. $Y$ are sorted in ascending order by using Eq. (7), in which $a(l)$ is the address index of the sorted chaotic sequence.

$$y_{a(l)} = ascend(y_l), \quad l = 1, 2, \cdots L \tag{7}$$

Denote $S$ as the scrambled signal, $S = \{s_l, 1 \le l \le L\}$, $s_l = x_{a(l)}$, $l = 1, 2, \cdots L$.

② Encryption: the encrypted data is denoted by $A = \{a_l, 1 \le l \le L\}$, where $a_l = s_l \times y_l$.
③ Frame and segmentation: We divide $A$ into $P$ non-overlapping frames and denote the $i$-th frame as $A_i$, which has $N$ samples. We cut $A_i$ into three non-overlapping segments, denoted by$A1_i$, $A2_i$ and$A3_i$. $A1_i$ has $n$ samples, and $A1_i(l)$ denotes the $l$-th sample of $A1_i$. $A2_i$ and$A3_i$ are divided into $M$ sub-segments, the length of each sub-segment is $m$. And the $j$-th sub-segment is denoted by $A2_{i,j}$ and$A3_{i,j}$, $1 \le j \le M$. The method of frame and segmentation is shown in Fig. 8.

④ Feature calculation: we denote $Y1 = \{y_l, 1 \le l \le m\}$. According to the Eq. (1), we calculate the RE features of $A2_{i,j}$ to $Y1$, and $A3_{i,j}$ to $Y1$, which are denoted by $E2_{i,j}$ and $E3_{i,j}$, respectively.

### 3.1.2 Watermark generation

Watermark bits are generated by the sequence $Y$ and frame number, containing two parts.

① The sequence $Y$ is divided into $P$ sets, and the $i$-th set is denoted by $Y_i=\{y_{(i-1)\times n+1}, y_{(i-1)\times n+2},\cdots y_{i\times n}\}$. Each set is mapped to one frame number. In this paper, the $i$-th set $Y_i$ is mapped to $i$, that is

$$\left\{y_{(i-1)\times n+1}, y_{(i-1)\times n+2}, \cdots y_{i\times n}\right\} \iff i \tag{8}$$

The set $\{y_{(i-1)\times n+1}, y_{(i-1)\times n+2},\cdots y_{i\times n}\}$ will be embedded into the segment $A1_i$, and as the identification of $i$-th frame.

② The frame number $i$ is converted into binary bits, denoted by $W_i=\{w_{i,j}|w_{i,j}\in(0,1), 1\le j\le M\}$, which is embedded into the segment $A2_i$ and $A3_i$.

### 3.1.3 Watermark embedding

The watermark bits generated above are embedded by using the following method.

① We embed $Y_i$ into $A1_i$. The method is that we substitute all the samples of $A1_i$ by using the elements of $Y_i=\{y_{(i-1)\times n+1}, y_{(i-1)\times n+2},\cdots y_{i\times n}\}$, expect for the value of first place after decimal point and the sign "+" or "-". An example is shown in Table 1, in which $n=5$, and $WA1_i(l)$ is $l$-th sample of the watermarked signal $WA1_i$.

② $W_i$ is embedded into the segment $A2_i$. For $w_{i,j}$, if $w_{i,j}=0$, it is embedded by using Eq. (9); and if $w_{i,j}=1$, it is embedded by using Eq. (10).

$$QE2_{i,j} = \begin{cases} \left\lfloor E2_{i,j}/\Delta \right\rfloor \times \Delta + \Delta/2, \left\lfloor E2_{i,j}/\Delta \right\rfloor \bmod 2 = 0 \\ \left(\left\lfloor E2_{i,j}/\Delta \right\rfloor - 1\right) \times \Delta + \Delta/2, \left\lfloor E2_{i,j}/\Delta \right\rfloor \bmod 2 = 1 \end{cases} \tag{9}$$

$$QE2_{i,j} = \begin{cases} \left\lfloor E2_{i,j}/\Delta \right\rfloor \times \Delta + \Delta/2, \left\lfloor E2_{i,j}/\Delta \right\rfloor \bmod 2 = 1 \\ \left(\left\lfloor E2_{i,j}/\Delta \right\rfloor + 1\right) \times \Delta + \Delta/2, \left\lfloor E2_{i,j}/\Delta \right\rfloor \bmod 2 = 0 \end{cases} \tag{10}$$

where $\Delta$ is the quantification step, and $QE2_{i,j}$ is the RE feature quantified. Denote $WA2_i=\{WA2_{i,j}, 1\le j\le M\}$ as the watermarked signal, based on the Eq. (5), $WA2_{i,j}(l)$ can be obtained by

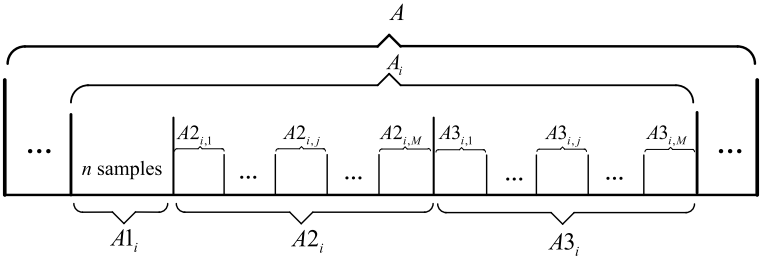



**Fig. 8** The frame and segment method

**Table 1** Example of watermark embedding

| $Y_i$ | $A1_i(l)$ | | $WA1_i(l)$ |
|---|---|---|---|
| 0.2637 | 0.4865 | → | 0.4637 |
| 0.6208 | 0.3494 | → | 0.3208 |
| 0.3080 | -0.1267 | → | -0.1080 |
| 0.0317 | 0.5523 | → | 0.5317 |
| 0.2039 | -0.0037 | → | -0.0039 |

$$WA2_{i,j}(l) = A2_{i,j}(l)\sqrt{10^{\frac{QE2_{i,j}-E2_{i,j}}{10}}} \tag{11}$$

where $A2_{i,j}(l)$ denotes the $l$-th sample of $A2_{i,j}$, $1 \le l \le m$.

By using the same method, $W_i$ is embedded into the segment $A3_i$, and the corresponding watermarked signal is denoted by $WA3_i$. $WA1_i$, $WA2_i$ and $WA3_i$ are concatenated to form the watermarked signal $WA_i$, $WA_i = WA1_i \| WA2_i \| WA3_i$, $1 \le i \le N$, which is the $i$-th frame of watermarked signal. The process of watermark generation and embedding is shown in Fig. 9.

### 3.2 Content authentication and tamper recovery

Suppose $WA = \{wa_l, 1 \le l \le WL\}$ is the watermarked signal in encrypted domain, downloaded from third-party servers, and $WL$ is the length of the signal. For the authentication method, if one frame is regarded as the original one, there are three conditions to be met, and the steps are as following.

#### 3.2.1 Preprocessing

① Frame and segmentation: using the method in Section 3.1, watermarked signal $WA$ is framed and segmented. Denote $WA_i$ as the $i$-th frame, $WA1_i$, $WA2_i$, and $WA3_i$ as the three segments of $WA_i$, and $WA1_i$ has $n$ samples. $WA2_{i,j}$ and $WA3_{i,j}$ represent the $j$-th sub-segment of $WA2_i$, and $WA3_i$, respectively, $1 \le j \le M$. The lengths of the segments are the same as those in Section 3.1.

② Features calculation: the RE feature of $WA2_{i,j}$ and $WA3_{i,j}$ to $B$ is calculated, and denoted by $\boldsymbol{WE}2_{i,j}$ and $\boldsymbol{WE}3_{i,j}$.

#### 3.2.2 Authentication

① According to the sequence generated by Eq. (6), we can get the $i$-th set $Y_i = \{y_{(i-1)\times n+1}, y_{(i-1)\times n+2}, \cdots y_{i\times n}\}$. If the values of $y_{(i-1)\times n+1}$, $y_{(i-1)\times n+2}$, $\cdots y_{i\times n}$ are the same as $WA1_i(l)$, $1 \le l \le n$, apart from the sign and the value of first place after decimal point, the segment $WA1_i$ is regarded to be original, where $WA1_i(l)$ represents the $l$-th sample of $WA1_i$. The relationship between $y_{(i-1)\times n+l}$ and $WA1_i(l)$ is denoted by

$$y_{(i-1)\times n+l} \iff WA1_i(l) 1 \le l \le n \tag{12}$$

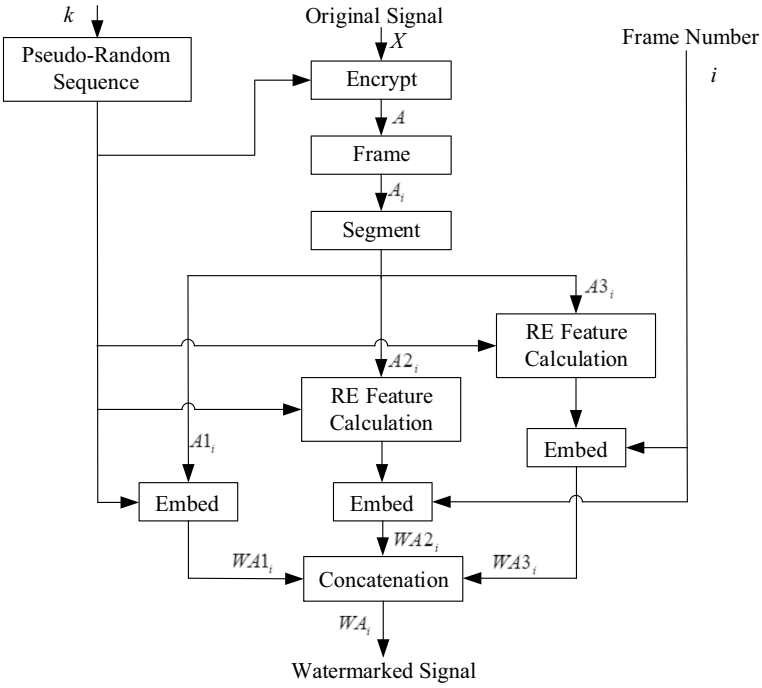



**Fig. 9** The process of watermark generation and embedding

② For the segment $WA2_i$and $WA3_i$, denote $W'_i = \left\{ w'_{i,j} \middle| w'_{i,j} \in (0,1), 1 \le j \le M \right\}$ as the watermark bit extracted from $WA2_i$, where $w'_{i,j} = \left\lfloor \boldsymbol{WE}2_{i,j}/\Delta \right\rfloor \bmod 2$. Similarly, for $WA3_i$, denote $W^*_i = \left\{ w^*_{i,j} \middle| w^*_{i,j} \in (0,1), 1 \le j \le M \right\}$ as the watermark bit extracted from $WA3_i$, and $w^*_{i,j} = \left\lfloor \boldsymbol{WE}3_{i,j}/\Delta \right\rfloor \bmod 2$.

③ Under the condition that $WA1_i$ is original, $W'_i = W^*_i$, and the frame number $i$ extracted from $WA2_i$or$WA3_i$matches the set $Y_i$ (shown in Eq. (8)) successfully, the $i$-th frame is regarded to be intact. Otherwise, it indicates that the $i$-th frame has been tampered. The process of watermark extraction and content authentication is shown in Fig. 10.

### 3.2.3 Tamper recovery

If $i$-th frame can not go through the authentication process successfully, move and authenticate next $N$ continuous samples, until we find $N$ continuous samples, denoted by$WA_{i'}$, authenticated successfully. Then, extract the frame number of the previous frame denoted by $WA_{i-1}$ that can pass the authentication, where $i-1$ identifies the frame immediately before the first frame failing the authentication process. From the difference of $i-1$ and $i'$, the tampered frames can be identified. The tamper location identification method is shown in Fig. 11.

Based on tamper location, the attacked frames are substituted by 0 amplitude signals. Then the samples are divided by the corresponding pseudo-random sequence, to perform decryption. The recovered signal can be obtained by anti-scrambling transformation.

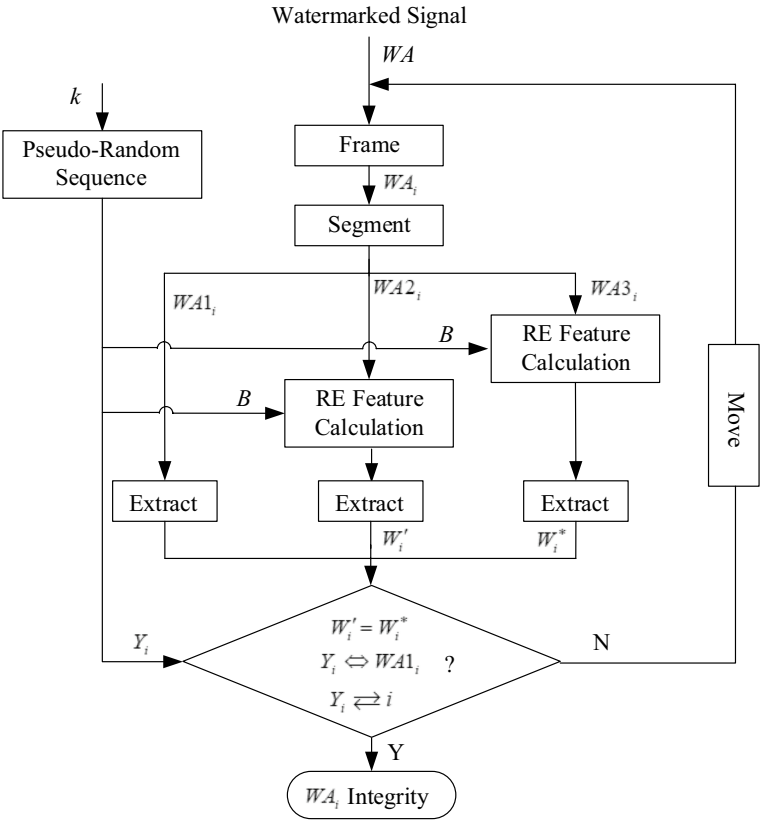



**Fig. 10** The process of watermark extraction and content authentication

## 4 Performance analysis and experimental results

The main goal of the proposed scheme is to protect the sensitive audio. In this Section, we recorded 100 audio as the test signal, including trade secrets and business meetings. All the signals are 16-bit quantified mono signals and sampled at 22.05 kHz. The parameters used are $L=401{,}000$, $P=20$, $M=10$, $N=20{,}050$, $n=50$, $k=0.287$, $\mu=3.9$ and the meaning of each parameter is shown in Section 3. The performance analysis and experimental results are shown in the following subsections.

### 4.1 Inaudibility and recovering capability

The inaudibility means that watermark embedded is inaudible, which reflects the change degree of original audio after watermarking. In this paper, subjective difference grades (SDG) [5] is used to test the inaudibility subjectively. The meaning of each score in SDG is listed in Table 2.

The average values of SDG for the 100 test signals at various quantification steps are tested, and the results are shown in Fig. 12. It can be seen that the SDG values are large

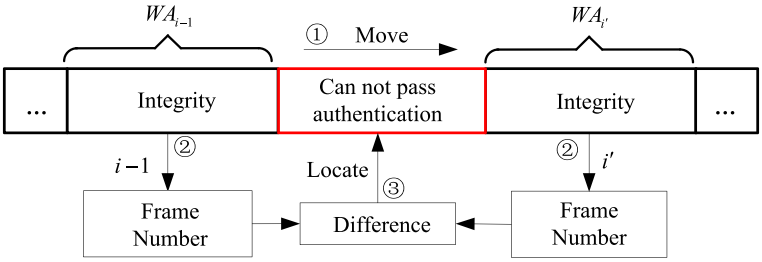



**Fig. 11** The method of tamper location identification

than $-1.5$, $0 < \Delta \leq 1$. Based on the score meaning in Table 2, it's concluded that the quality of the audio is not affected by watermark embedding.

The average SDG values of the recovery signal for the 100 test signals with different number of samples deleted are shown in Fig. 13, in which ND represents the number of samples deleted, and the vertical coordinate represents SDG value of the corresponding recovery signal. The results shown in Fig. 13 are acquired from 10 listeners. It's concluded that the SDG values decrease as the number of samples deleted increases, and the quality of the recovered signal is poorer when the attack degree is stronger.

It's found that, when the SDG value is greater than $-1.5$, the meaning of the recovered signal is easy to be acquired by listeners, and has the same expressed meaning as the original signal. Figure 14 shows the maximum number of deletion samples allowed for the 100 audio signals, under the condition that the SDG value is greater than $-1.5$. In Fig. 14, ND represents the maximum number of samples that can be deleted, and NT(h) represents the $h$-th test signal, $1 \leq h \leq 100$.

In this paper, we denote $R_c$ as the recovery capability, and the definition is shown in Eq. (13):

$$R_c = \frac{L_c}{L} \tag{13}$$

where $L_c$ represents the maximum number of samples that can be deleted under the condition that the SDG value is greater than $-1.5$, and $L$ represents the length of original signal.

According to the SDG values listed in Table 2, and based on the results shown in Figs. 13 and 14, it can be seen that the maximum number of the deletion samples allowed is about 50,000 under the condition that the SDG value is greater than $-1.5$. In addition, the recovery capability of the proposed scheme is about 1/8.

**Table 2** Subjective Difference Grades

| SDG | Description of impairments | Quality |
|---|---|---|
| 0 | Imperceptible | Excellent |
| -1 | Perceptible, but not annoying | Good |
| −2 | Slightly annoying | Fair |
| −3 | Annoying | Poor |
| −4 | Very annoying | Bad |

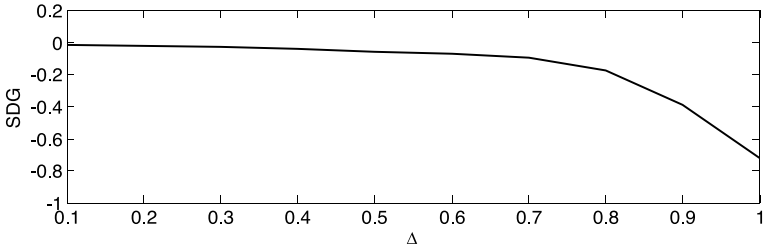



**Fig. 12** The average values of SDG under various quantization steps

## 4.2 Security

### 4.2.1 Encrypted signal

For the encryption method proposed in this paper, the audio is scrambled first, and then the scrambled audio is multiplied by a pseudo-random sequence. On the one hand, the scrambled audio is very close to a noise signal. Thus, it's difficult to acquire the expressed meaning from the scrambled signal. On the other hand, the statistical characteristics of original audio is changed by multiplication, which makes it more difficult to successfully carry out attacks that are based on statistical features.

Figure 15 shows the original signal and the corresponding histogram. The encrypted signal is shown in Fig. 16a, and the histogram is shown in Fig. 16b. By comparing with the result shown in Figs. 15b and 16b, it can be seen that the histogram of encrypted signal is different from that of the original one. This is because that the statistical characteristics of original signal are changed after encryption.

From the analysis and the test results, it's concluded that the encrypted audio is different from the original one, and it's very hard for attackers to get the expressed meaning from the encrypted audio. So, the encryption method is effective for privacy protection.

### 4.2.2 Embedding method

For digital watermark schemes, if watermark bit embedding is based on public features, the scheme is vulnerable to substitution attack [5]. For the scheme proposed in this paper, the watermark embedding contains two components, pseudo-random sequence and frame

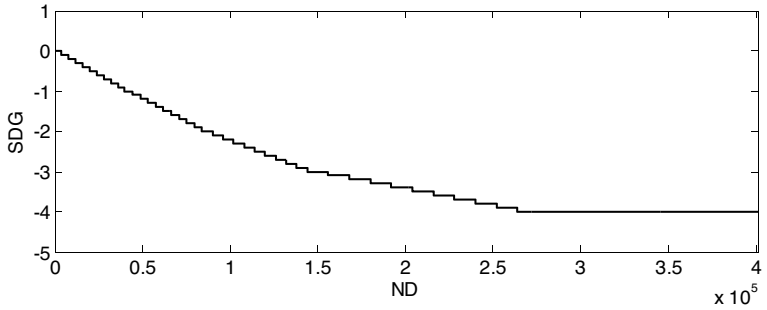



**Fig. 13** The SDG values of the recovery signal for different number of samples deleted

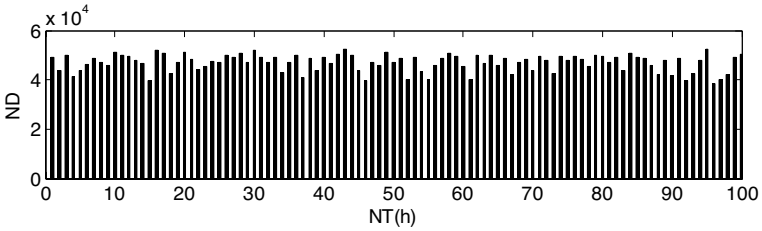



**Fig. 14** The maximum number of samples can be deleted under the condition that SDG greater than −1.5

number. In the following subsections, the security features of the two components of the embedding method are analyzed respectively.

① For pseudo-random sequence embedding.

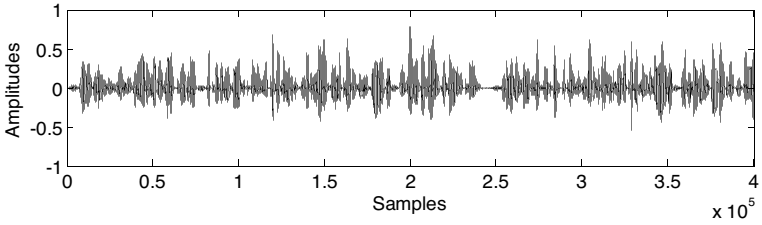
In this paper, the pseudo-random sequence generated by Eq. (6) is divided into different sets, and the $i$-th set is mapped to the frame number $i$. If the pseudo-random sequence embedded into the first part $A1_i$ of $i$-th frame is attacked, the values extracted from $A1_i$ will be different from the $i$-th set. So, it can be detected with high probability. The probability that the attack can not be detected is $1/10^{3n}$. Thus, the ability against the attack is

$$A_p = 1 - \frac{1}{10^{3n}} \tag{14}$$

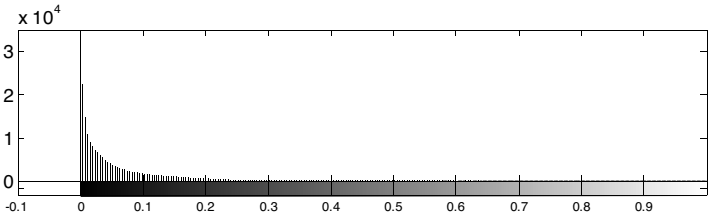
where $A_p$ represents the ability against the attack for one frame, $n$ represents the length of $A1_i$.

② For frame number embedding.

The frame number embedding is performed by quantifying the RE feature. Based on the analysis in Section 2.1, it can be seen that it's difficult to obtain the RE feature without the

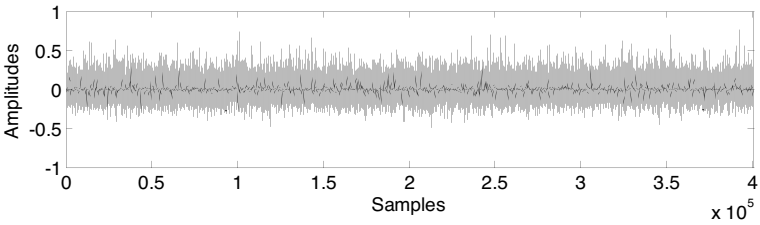



(a) The original audio signal

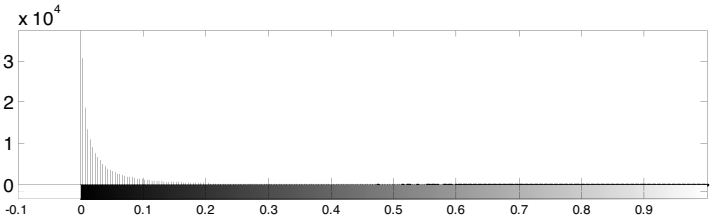



(b) Histogram of original audio signal

**Fig. 15** The original audio and the histogram




(a) The encrypted signal




(b) Histogram of encrypted signal

**Fig. 16** The encrypted audio and the histogram

signal $Y$. In this paper, the signal $Y$ is regarded as noise signal generated by logistic chaotic mapping, which is related to the secret key. So, for attackers without the secret key, it's hard to get the feature. If attackers select a secret key randomly to perform the substitution attack, the probability that watermark bits can be extracted from the second are equal to those from the third segment is $1/2^M$. That is, for the watermark embedding method proposed, the ability against the substitution attack [5] for one frame is

$$A_s = 1 - \frac{1}{2^M} \tag{15}$$

where $A_s$ represents the ability against substitution attack, and $M$ represents the length of watermark bits.

Based on the performance analysis above, the abilities of the scheme are compared with those proposed in [2, 9–12], shown in Table 3, in which Ability I represents the ability resisting to substitution attack, and Ability II represents the security of watermark embedding. From the comparison results, it's concluded that the scheme has the ability resisting to substitution attack, and can improve the security of watermark embedding.

### 4.3 Precise tamper location identification capability

For the encryption method proposed, the first step is to scramble host audio signal. In order to obtain the original signal, there must be an anti-scrambling process. If the encrypted signal is subjected to de-synchronous attack, e.g. deletion attack or insertion attack, the signal obtained by anti-scrambling is different from the original one to a great extent. So it's very desirable for the scheme proposed to locate the attacked signals precisely.

**Table 3** Ability comparison for different schemes

| Scheme | Features Used | Ability I | Ability II |
|---|---|---|---|
| [1] | Global feature (public) | No | No |
| [4] | High-order difference statistics (public) | No | No |
| [3] | Histogram shape (public) | No | No |
| [2, 10, 12] | Energy (public) | No | No |
| [9] | Feature points (public) | No | No |
| [11] | DYWT (public) | No | No |
| Proposed | Relative Energy (secret) | Yes | Yes |

The method of tamper location in this paper is mainly based on the set obtained by pseudo-random sequence, shown in Fig. 17. Based on Eq. (14), we get that the probability of identifying tamper location precisely is $A_p$. Under the condition that $n=10$ in this section, the value of $A_p$ is very close to 1. So, the method can locate the watermarked and attacked frames precisely. Just based on the precise tamper location, if the attacked frames are substituted by using 0 amplitude signals, the signal closely to original one can be obtained by inverse scrambling.

### 4.4 Tamper recovery and analysis

We select one audio randomly and show the signal and the watermarked signal in Figs. 18 and 19, respectively. Then we delete 50,000 samples of the watermarked signal, which is shown in Fig. 20. Based on the proposed, we can get that the 3-th to 5-th frame are attacked, shown in Fig. 21, in which FN represents the frame number extracted, and $V=1$ represents that the corresponding frame is intact. Based on the tamper location result, substitute the attacked frames using 0 amplitude signals, which is shown in Fig. 22, and samples of the signal are divided by the pseudo-random sequence $Y=\{y_l, 1\le l\le L\}$. Then the tamper recovery signal can be obtained by anti-scrambling operation, and the result is shown in Fig. 23.

The Figs. 18, 19, 20, 21, 22 and 23 are examples of tamper location identification and recovery for attacks of different types. The SDG value of the recovered signal is −0.72. It can be seen that the scheme proposed has the ability of tamper location identification and

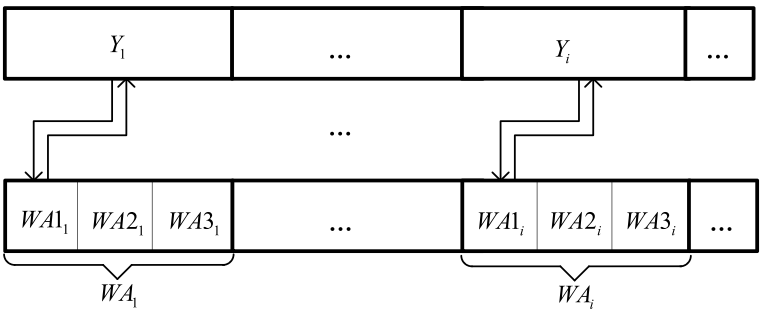



**Fig. 17** The method of tamper location precisely

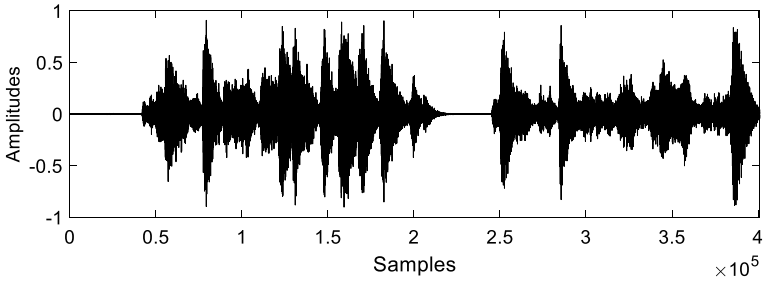



**Fig. 18** The test audio selected

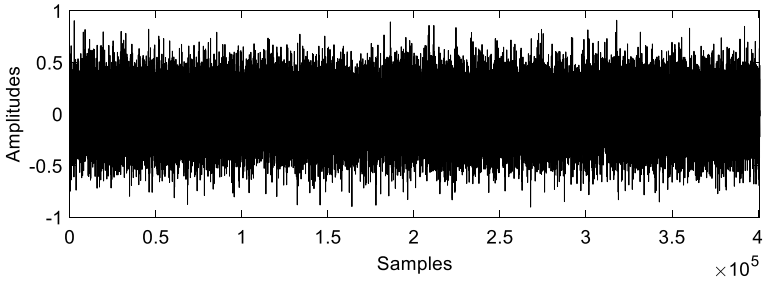



**Fig. 19** The watermarked data

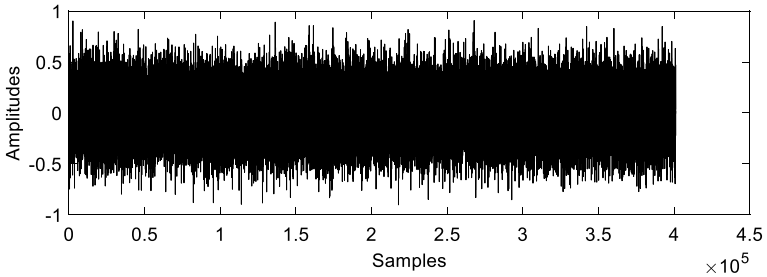



**Fig. 20** The attacked data

recovery for the encrypted audio, and the recovered signal is close to the original one. For other attacks, the steps of tamper recovery are similar.

It's worth noting that, for encrypted signal, the attacked content is substituted by using 0 amplitude signals. The reason is that, firstly, the substitution can make the attacked signal to be synchronized, which is very important for the scrambled signal. If the signal is de-synchronized, the signal obtained by anti-scrambling operation will be different from the original one. Secondly, 0 amplitude signals are meaningless signal, and they are spread to different parts after anti-scrambling and do not influence the expressed meaning of original signal. So the 0 amplitude signals are selected to substitute the attacked frame. And experiment results also show that the recovery performance by using 0 amplitude signals is better than other signals.

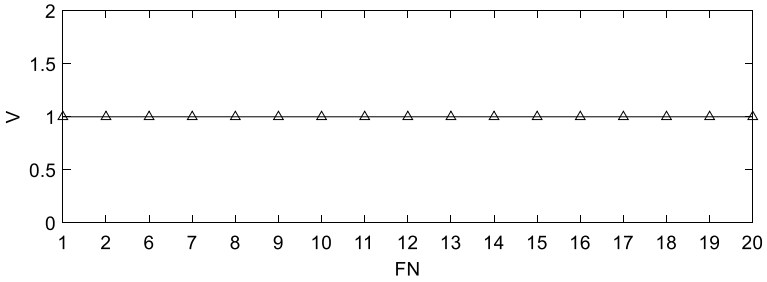



**Fig. 21** The tamper detection results

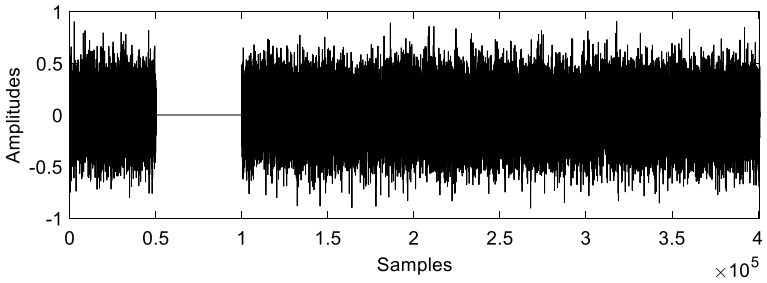



**Fig. 22** The data substituted by 0 amplitude signals

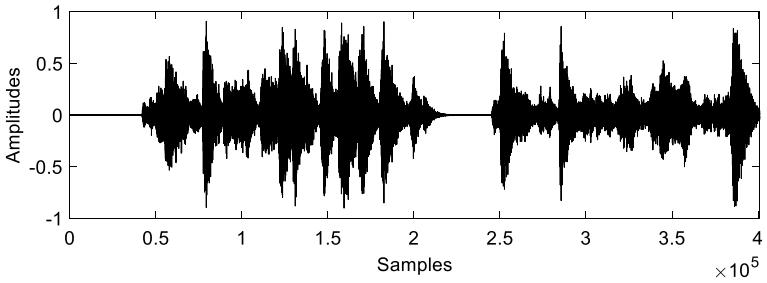



**Fig. 23** Tamper recovery result

## 5 Conclusion

Based on digital watermark, a content authentication and tamper recovery algorithm for encrypted audio signal is proposed. In the paper, a watermark embedding method based on signal RE feature and an encryption method for digital audio are proposed. It's hard for attackers to get meaningful content from the signal encrypted by using the method, which can protect the privacy of audio stored on third-party servers. The set of pseudo-random sequence embedded into watermarked signal is used to locate the watermarked and attacked frames. Based on the method, the attacked frames can be located precisely. Attacked frames are substituted by using 0 amplitude signals to perform tamper recovery. The scheme improves the security of watermark system, and has a certain ability of tamper recovery for encrypted audio signals. Experimental results show that the tamper recovery scheme proposed is effective.

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## Declarations

**Conflict of interest** The authors confirm that the manuscript has been submitted solely to this journal and is not published, in press, or submitted elsewhere. There are no financial interests or benefits dependent on this manuscript. There is also no conflict of interest with any other research.

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